

Andrés Jauch

BSc Data Science Student

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EDUCATION

BSc Data Science - FHNW University of Applied Sciences

Brugg-Windisch, CH

2024 – Present

- Focus on applied machine learning, statistical modeling, computer vision, and distributed systems.
- Core coursework: Linear Algebra, Probability & Statistics, Predictive Modeling, Software Engineering.

Commercial Apprenticeship & Federal Matura (BM1) - Wirtschaftsschule

Basel, CH

2020 – 2023

- Dual education combining business practice with advanced accounting, economics, and contract law.
- Graduated with Swiss Federal VET Diploma (EFZ) alongside academic baccalaureate (BM).

SELECTED PROJECTS

Rhine Swimability Monte Carlo Simulation | Python, NumPy, SciPy, Pandas

GitHub ↗

- Processed 23 years of sensor time series to model swimmable summer conditions in the Basel Rhine.
- Implemented MLE parameter estimation, bivariate normal sampling with Cholesky decomposition ($\rho \approx 0.50 - 0.57$), and a 10'000-run Monte Carlo simulation.
- Engineered a rolling-window (± 20 days) continuous day-checker avoiding discrete month-boundary jumps.

Tennis 3D Tracking & Analysis | Python, Computer Vision, Stereo Camera, ZED SDK, Streamlit

GitHub ↗

- Built an end-to-end stereo computer vision pipeline for real-time 3D ball and player position tracking.
- Calculated trajectory curves, court bounce detection, and velocity metrics served via an interactive dashboard.

Brain Health Challenge | Python, Scikit-Learn, Random Forest, Jupyter

GitHub ↗

- Biomedical challenge predicting neurological clinical outcomes from clinical health indicators.
- Implemented and evaluated Random Forest classification baselines and feature importance pipelines.

EXPERIENCE

Legal & Operations Assistant - Advokatur & Rechtsberatung TRIAS AG

Basel, CH

2023 – 2024

- Structured and digitized physical case files into centralized, searchable document archives.
- Automated routine clerical workflows and coordinated deadline-sensitive legal documentation.
- Monitored statutory court deadlines and handled formal correspondence with authorities and clients.

TECHNICAL SKILLS

Languages:

Databases & Data Handling:

Data Science & ML:

Platforms & Tools:

Spoken Languages:

Python, SQL, R, TypeScript, JavaScript, HTML/CSS, Bash
PostgreSQL, SQL Queries, Data Cleaning & Preprocessing (Pandas)
NumPy, SciPy, Pandas, Scikit-Learn, Matplotlib, Seaborn, Streamlit
Git, GitHub, Docker, Linux, Vite, Jupyter
German (Native), Spanish (Native), English (Fluent)